A Data-Driven Semantic Search of HKTVmall Products based on Product Characteristics

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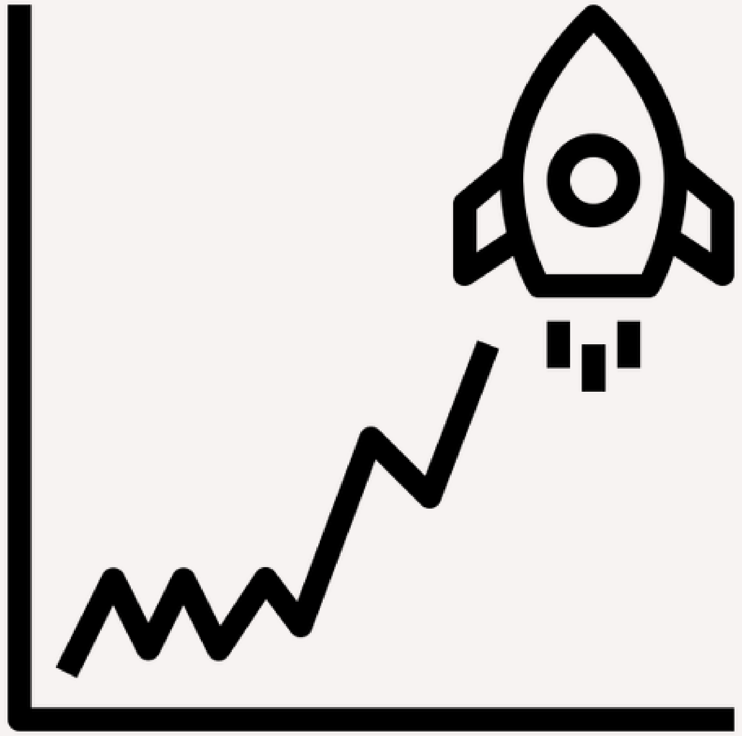
INTRODUCTION

Over the years, E-Commerce in HK has grown a lot

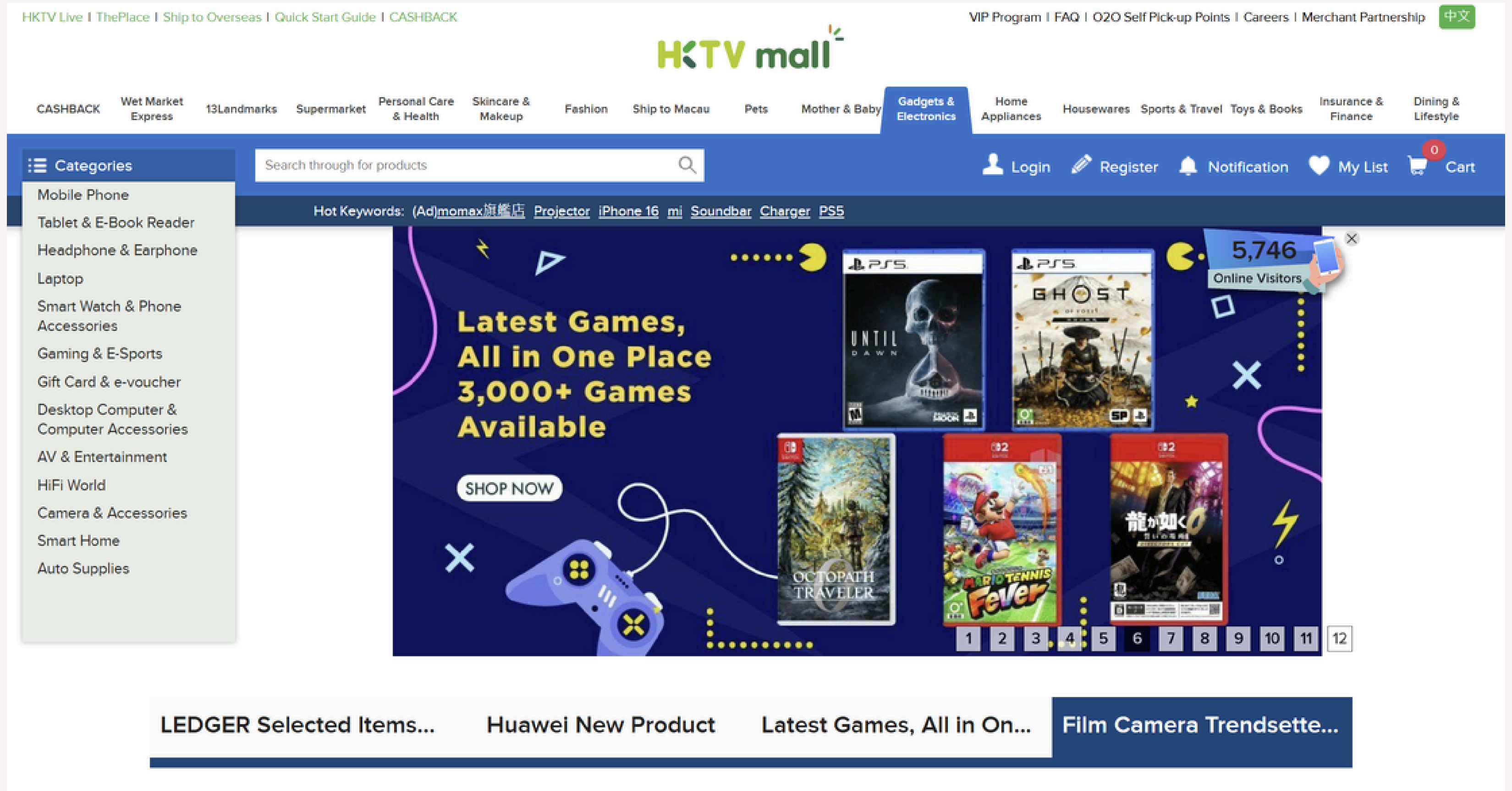
By 2027

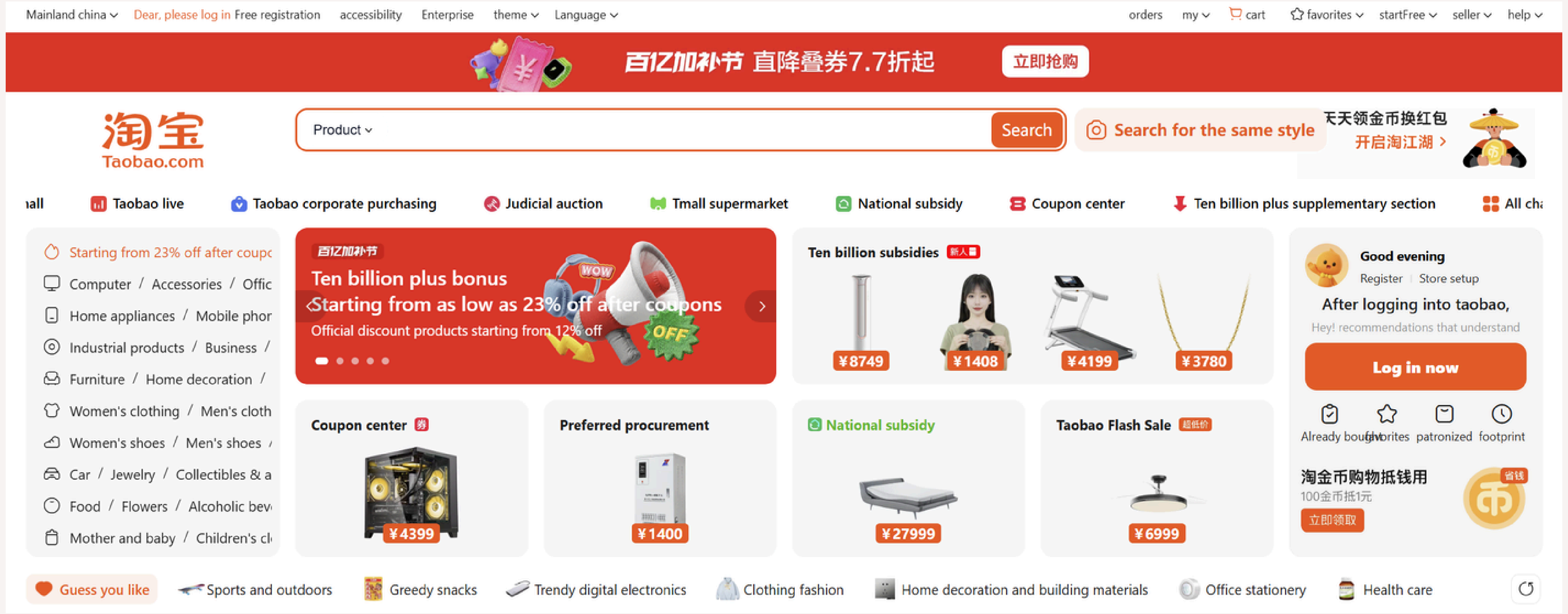


Estimated market volume
of US$ 30.5 billion

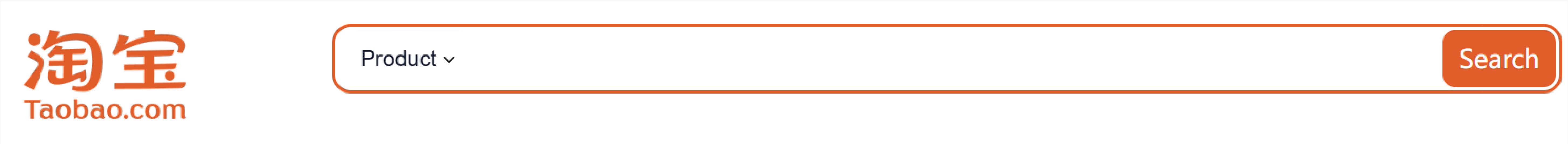
Payment and Commerce Market Intelligence. (2024). The Evolution of Hong Kong's E-Commerce Market.

Various e-commerce marketplace websites have emerged over the years





Most websites have a search bar to help find products



Search bar usually uses a form of TF-IDF/Keyword Search

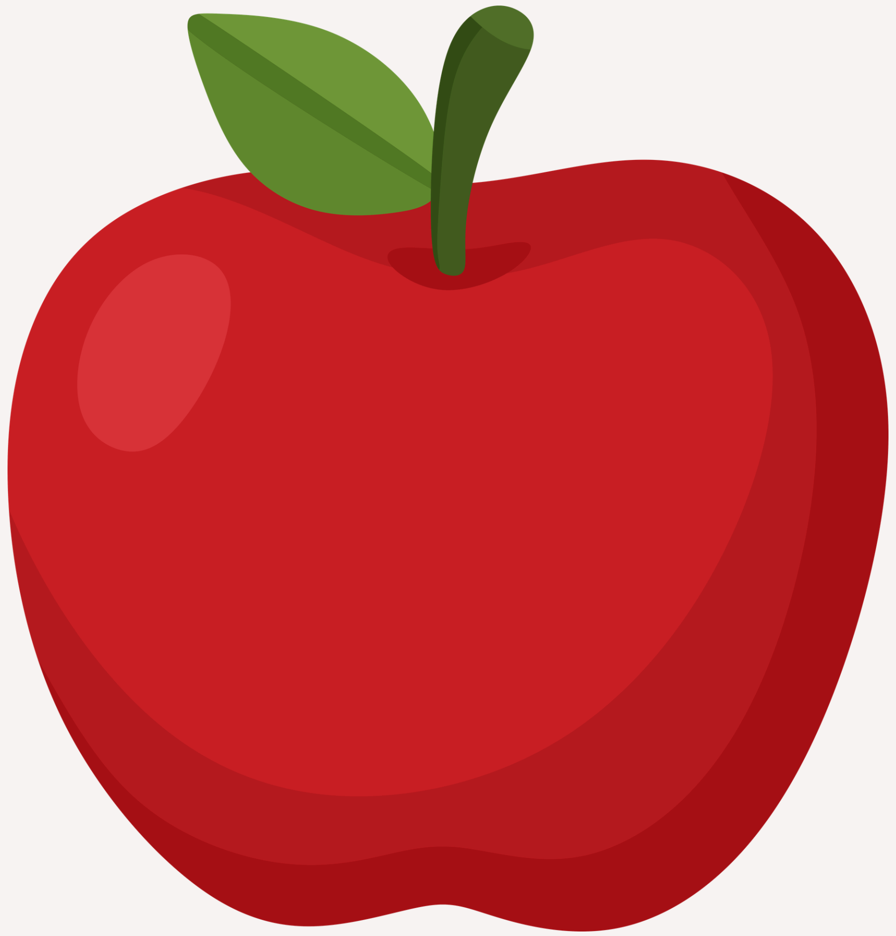
TF-IDF (term frequency-inverse document frequency)

$$TF = \frac{Number\ of\ times\ a\ word\ "X"\ appears\ in\ a\ Document}{Number\ of\ words\ present\ in\ a\ Document}$$

$$IDF = log\left(\frac{Number\ of\ Documents\ present\ in\ a\ Corpus}{Number\ of\ Documents\ where\ word\ "X"\ has\ appeared}\right)$$

$$TF\ IDF = TF * IDF$$

TF-IDF does not work with the meaning behind words



A possible solution is semantic search

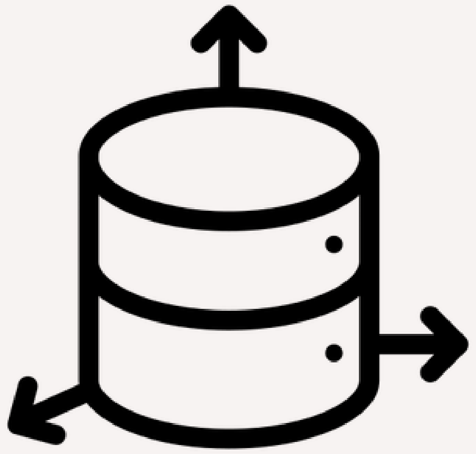
- As the name implies, semantic search uses the semantic of the word
- Pays attention to context
- Uses more advanced NLP techniques

METHODOLOGY

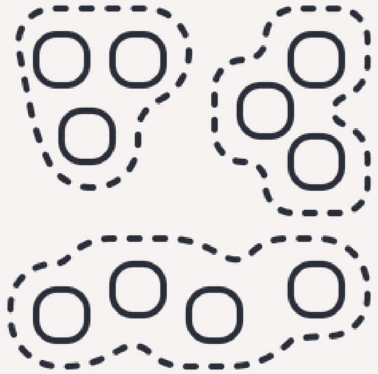
Pipeline

Data Collection

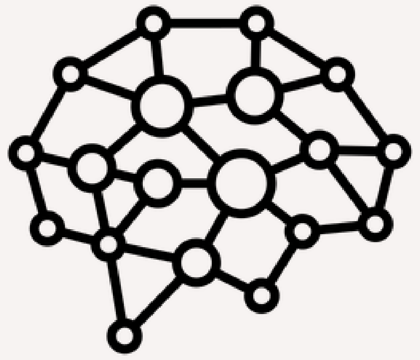
Preprocesing



Embeddings



Clustering



Semantic Search Model

Evaluation

Platform Setup

Data Collection through HKTVmall API

20
Keywords

20,000
Products Collected

38
Fields per Product

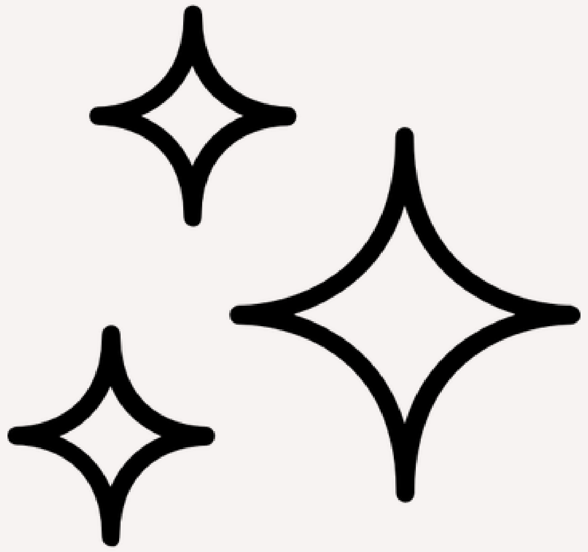
1,125
Unique Stores

via HKTVmall API

Text Preprocessing

Obtain English and Chinese text

Remove stopwords

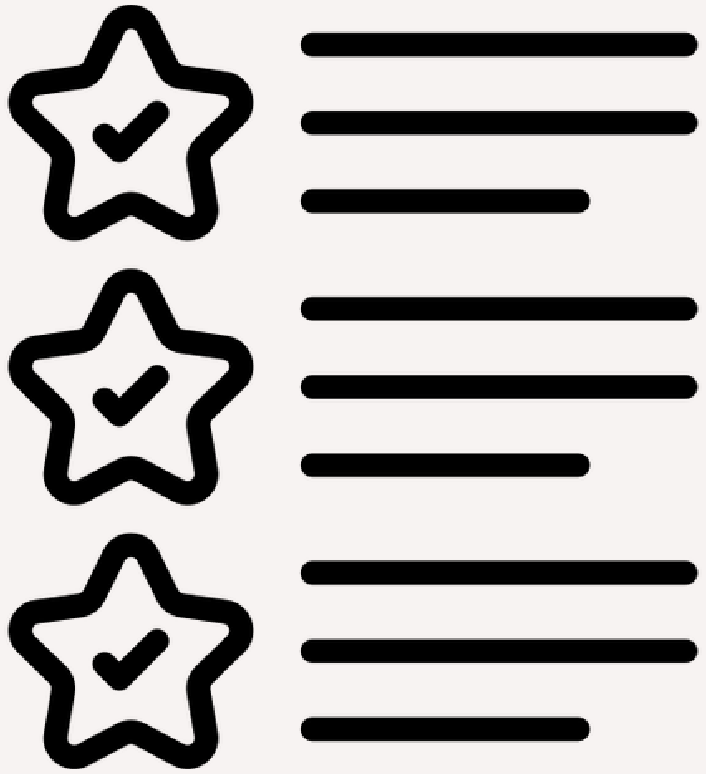


Clean/Normalize text by:

- Keep lowercase text and numbers
- Remove whitespace

Text Preprocessing

Tokenize text into individual words



Obtain features from the text

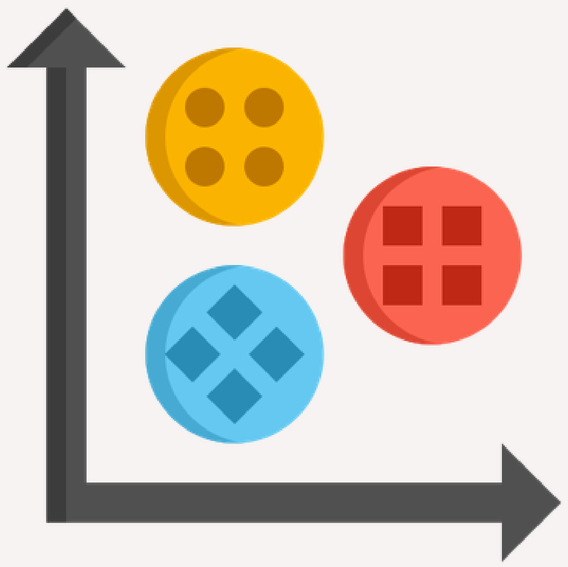
Text Embeddings

Obtain name and description of product from processed text

Use hugging face sentence transformer to obtain embeddings (paraphrase-multilingual-mpnet-base-v2)

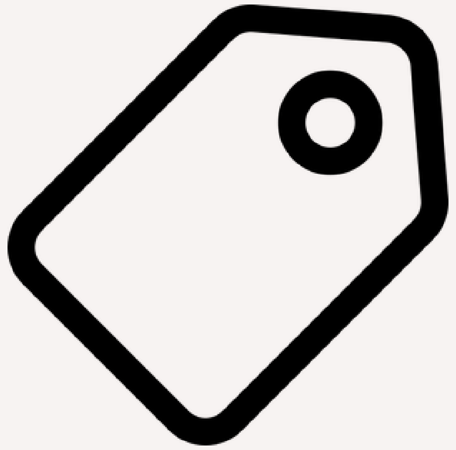
Visualize the embeddings

Clustering



Perform K-Means clustering on each keyword based on embedding value

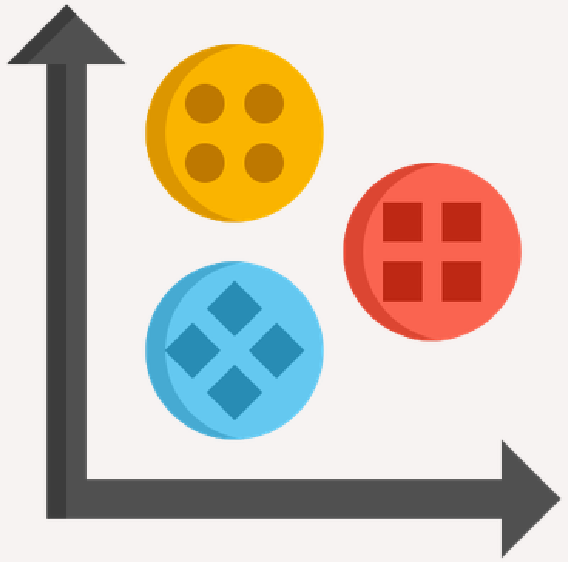
Compute Silhouette Score



Label the clusters into a readable form using TF-IDF

Semantic Search Model

Users enter a keyword



Results will be shown based on cluster labels

Cosine similarity will be used for more specific searches

Evaluation

Run both semantic search and keyword search on test sets

Calculate cluster overlap ratio

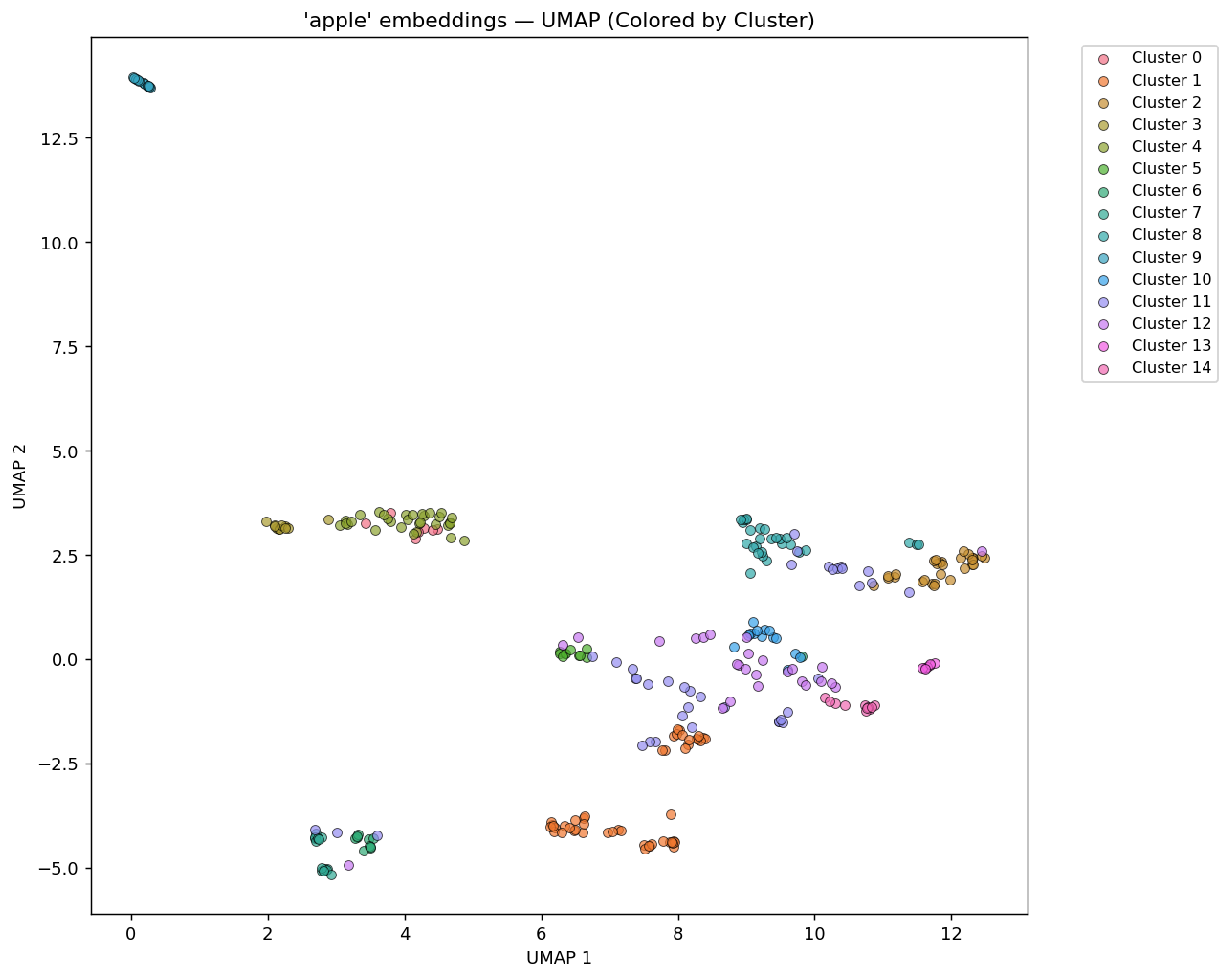
Average cosine score (Semantic search)

Average TF-IDF score (Keyword search)

RESULTS

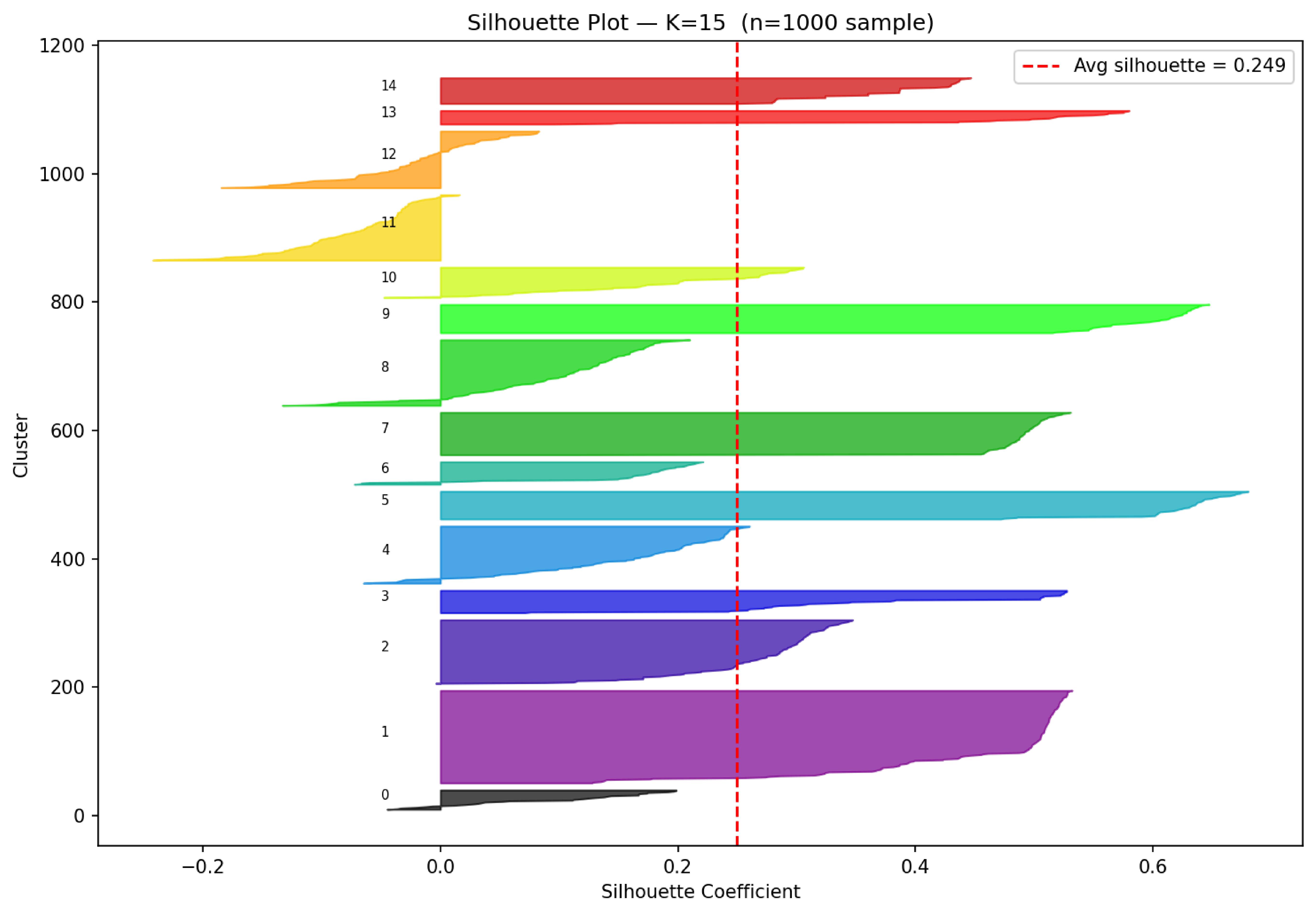
Semantic Product Clusters

(apple)



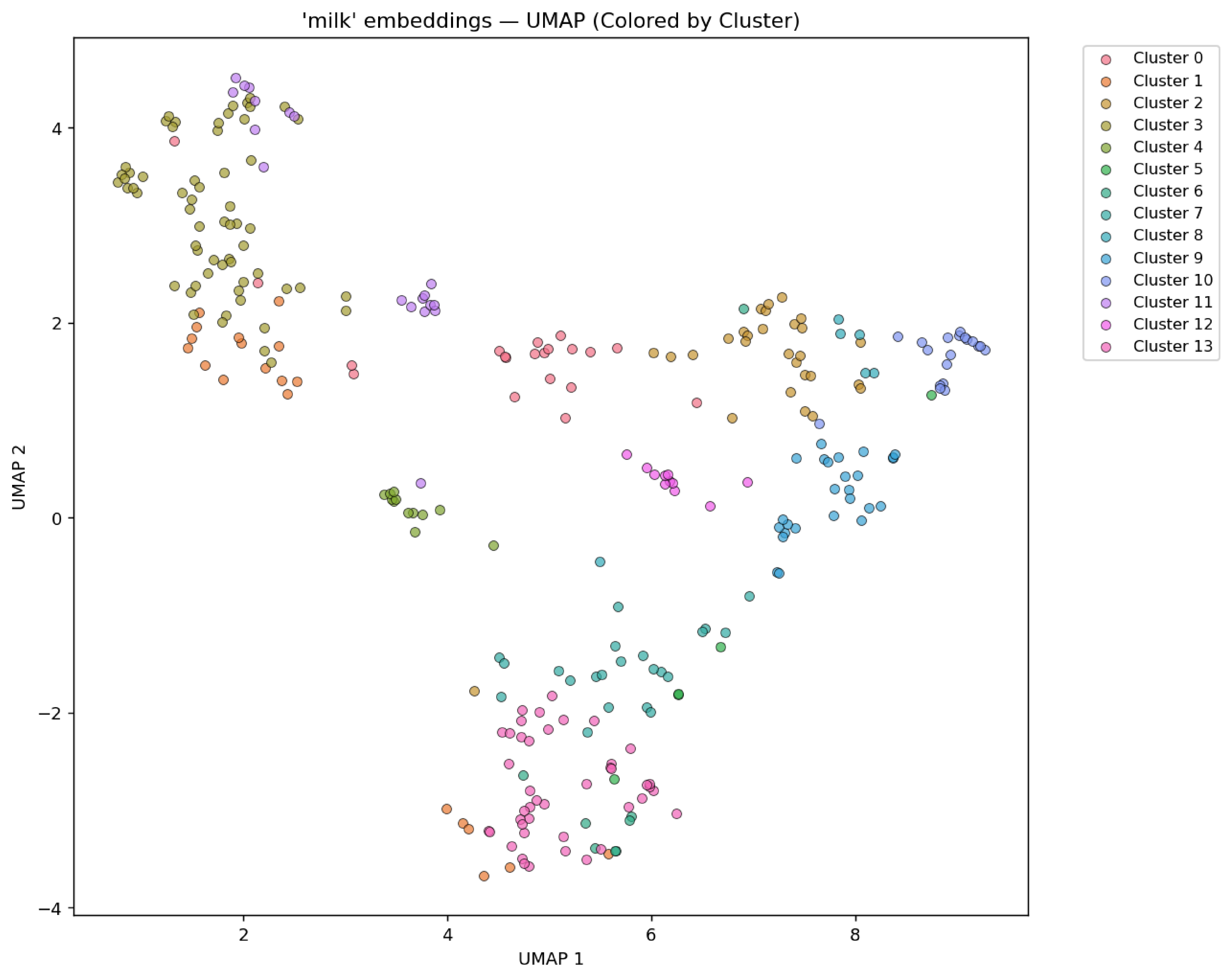
Semantic Product Clusters

(apple)



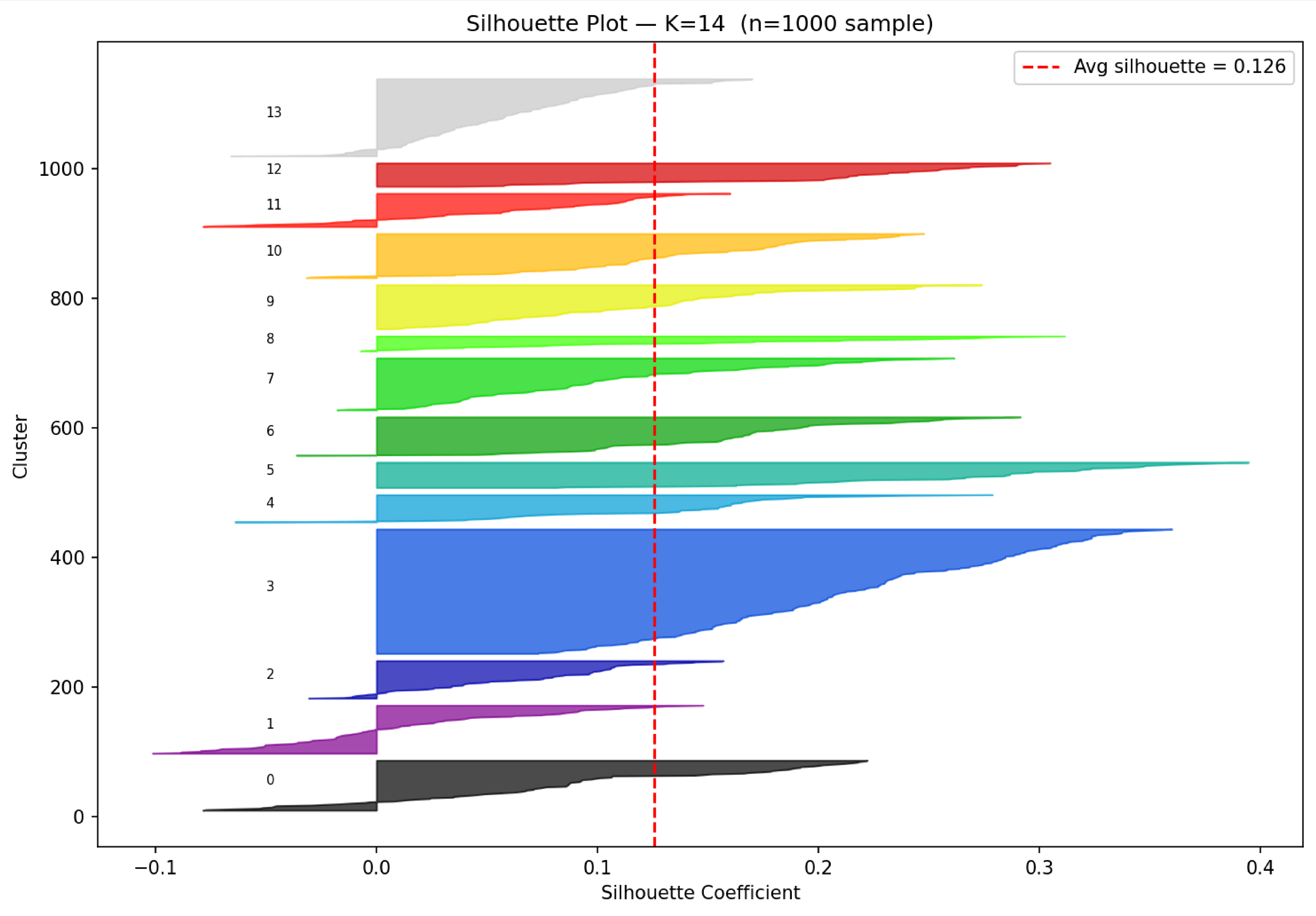
Semantic Product Clusters

(milk)

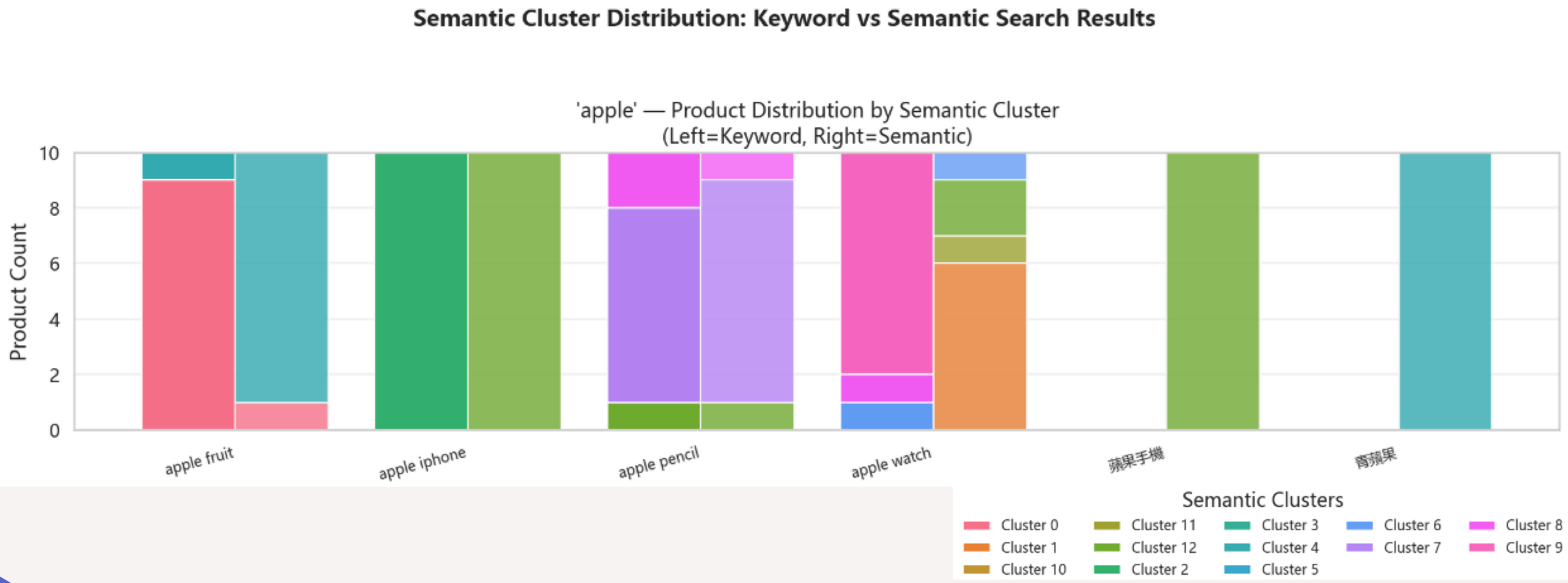


Semantic Product Clusters

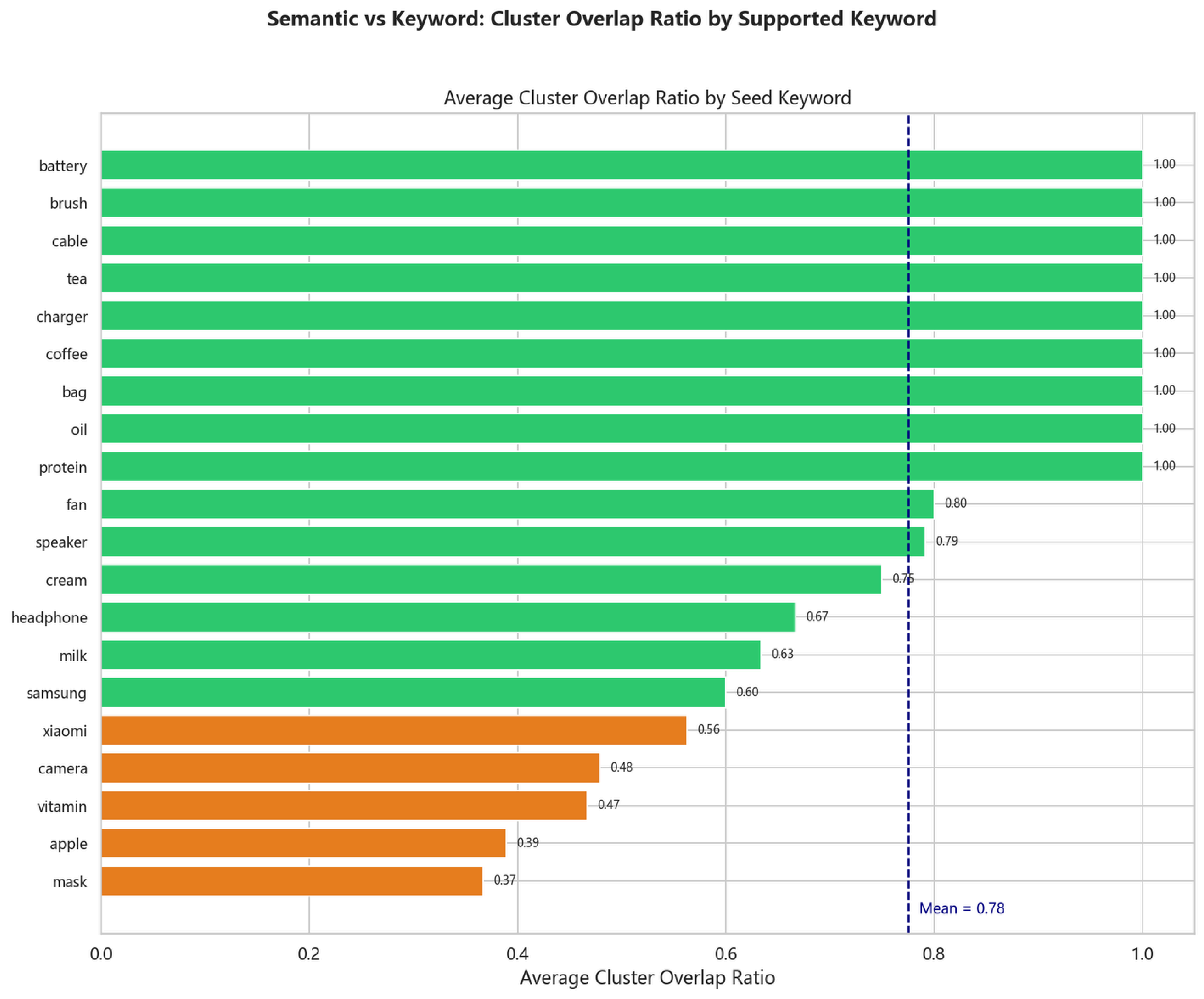
(milk)



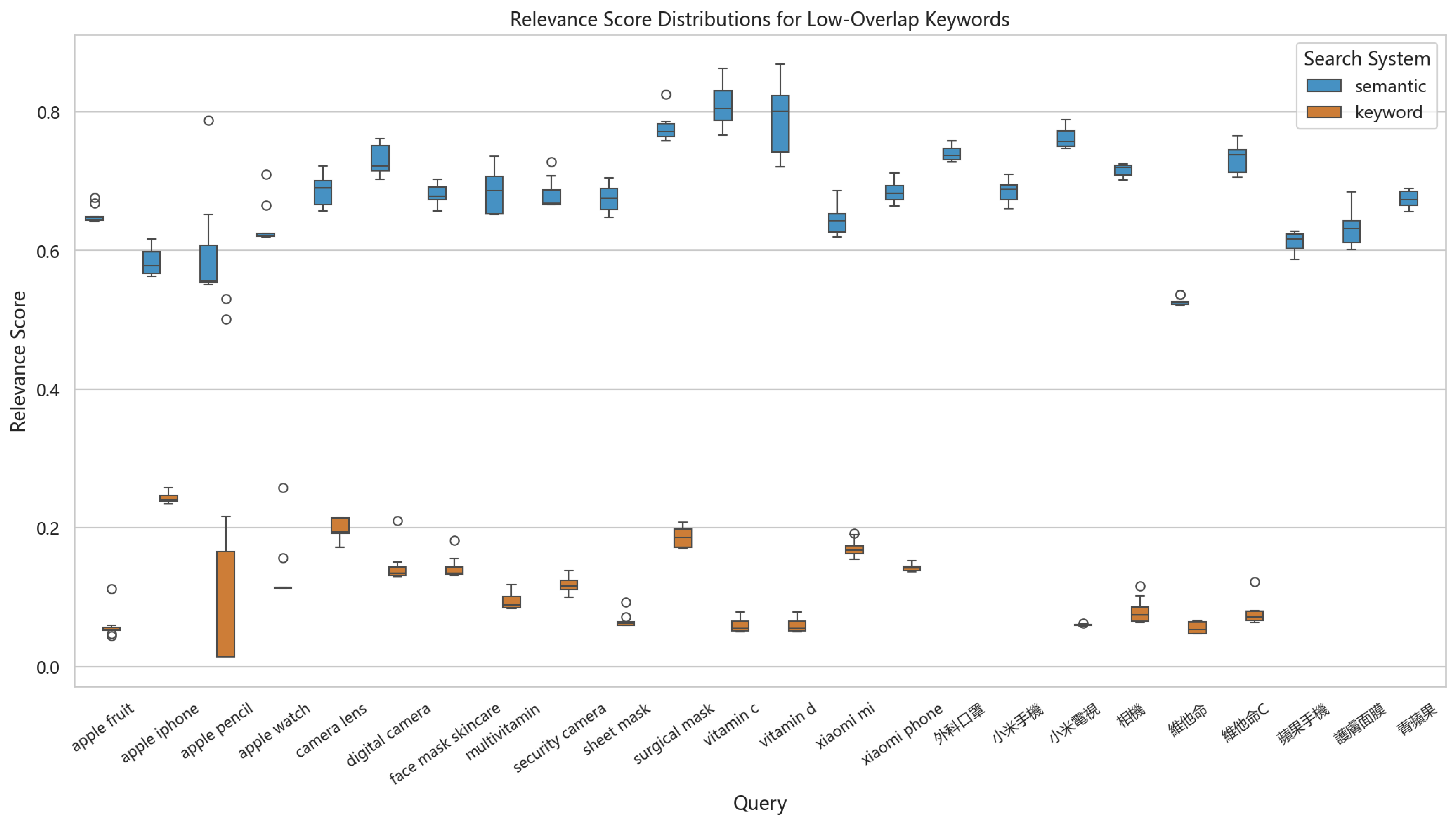
Disambiguated Search Groups



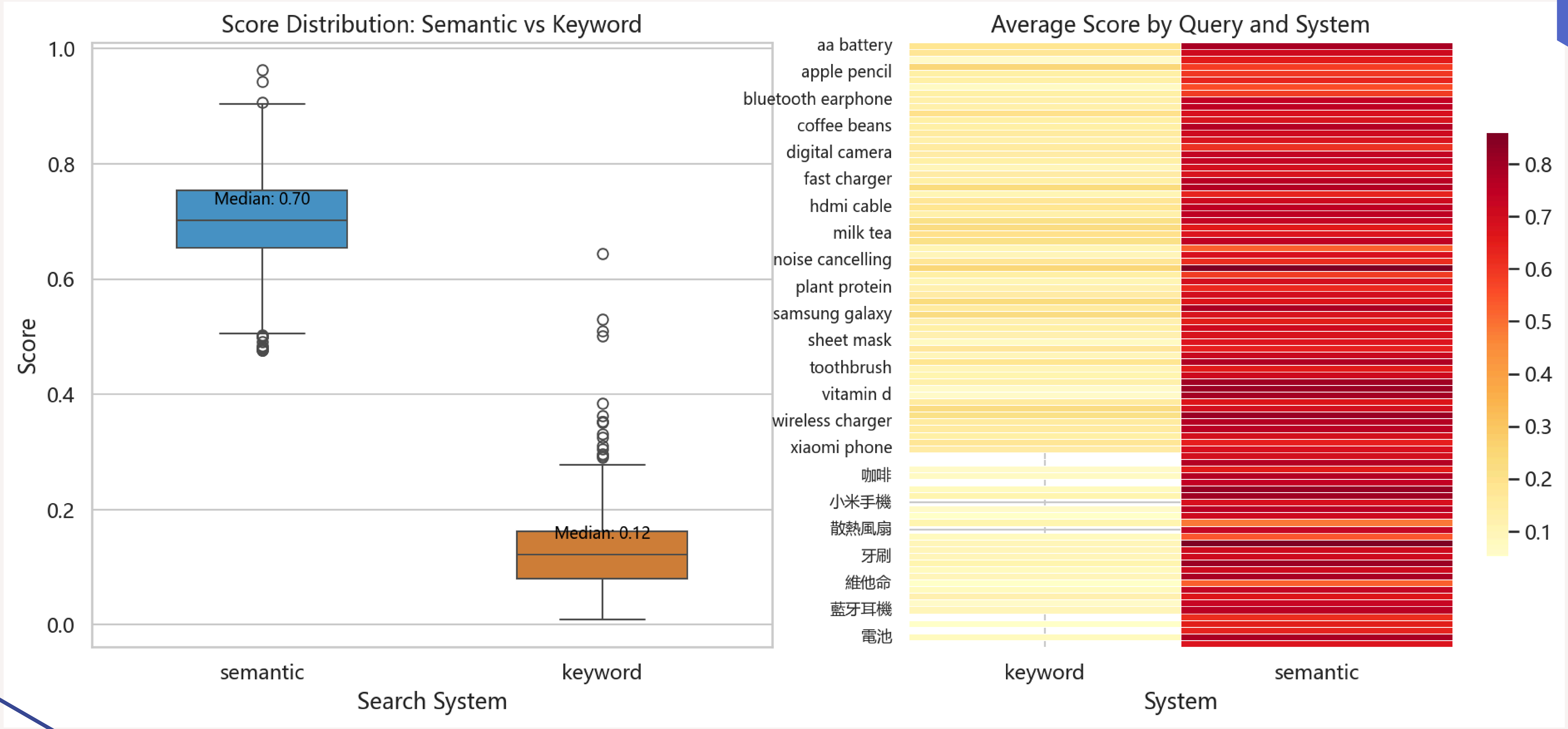
Disambiguated Search Groups



Disambiguated Search Groups



Quality-Ranked Product Lists per Semantic Meaning



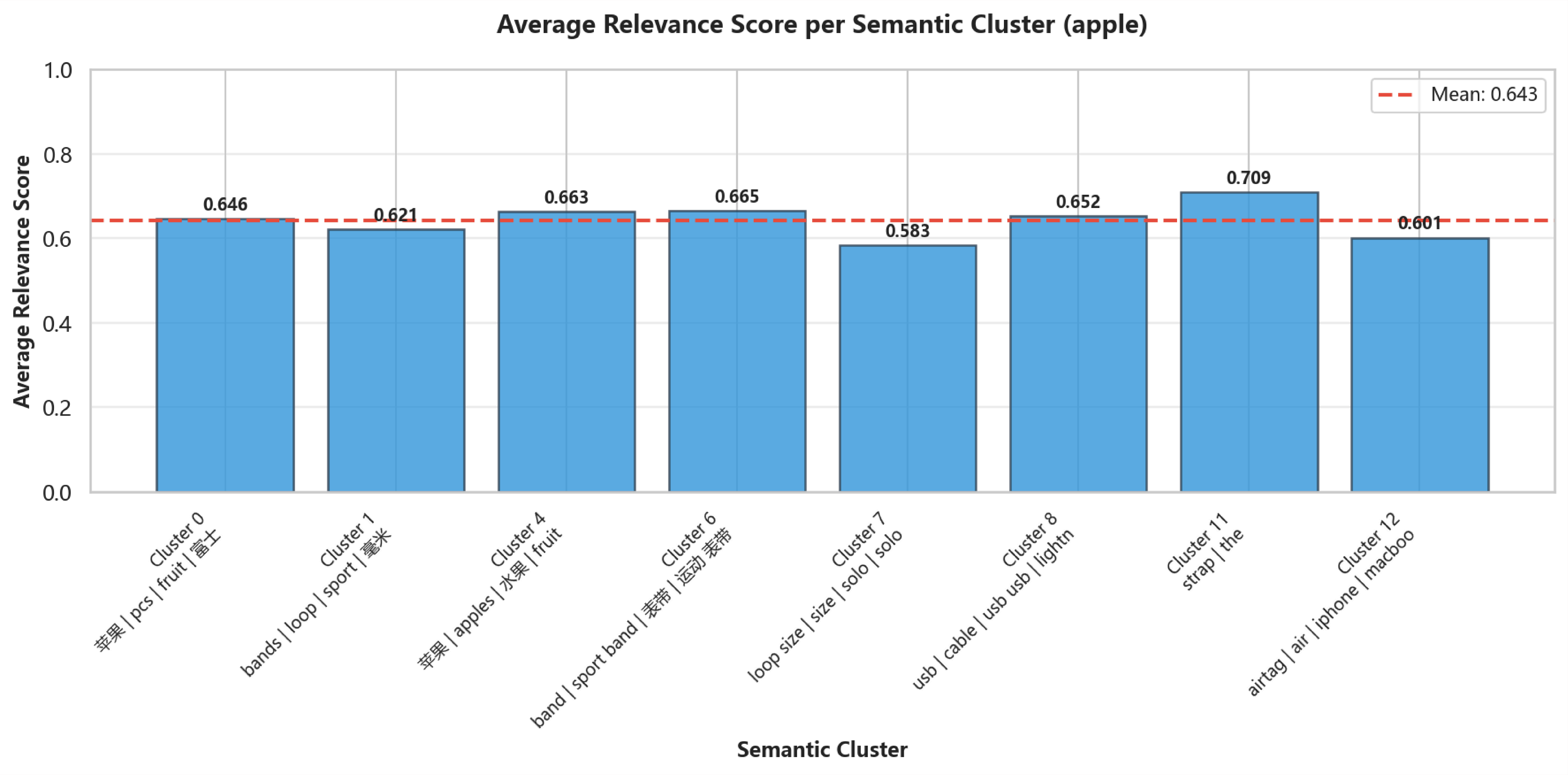
Quality-Ranked Product Lists per Semantic Meaning

cluster	cluster_label	rank	name_en	name_zh	**score**	query_with_max_score
0	苹果 | pcs | fruit | 富士	1	【8pcs】New Zealand / Can	【8個】紐西蘭/加拿大/小蘋果	0.646175	apple fruit
1	bands | loop | sport | 毫米	1	42mm Neon Yellow Solo Lo	Apple 42 毫米霓虹黃色單圈手	0.62469	apple watch
1	bands | loop | sport | 毫米	2	42mm Neon Yellow Solo Lo	Apple 42 毫米霓虹黃色單圈手	0.621193	apple watch
1	bands | loop | sport | 毫米	3	42mm Neon Yellow Solo Lo	Apple 42 毫米霓虹黃色單圈手	0.620959	apple watch
4	苹果 | apples | 水果 | fruit	1	Pollution-Free, Fresh Delive	無公害種植新鮮直送 8個裝 蘋	0.688911	青蘋果
4	苹果 | apples | 水果 | fruit	2	Pollution-Free, Fresh Delive	無公害種植新鮮直送 8個裝 蘋	0.688911	青蘋果
4	苹果 | apples | 水果 | fruit	3	Pollution-Free, Fresh Delive	無公害種植新鮮直送 5個裝 蘋	0.685076	青蘋果
6	band | sport band | 表带 | 运	1	WATCH S11 GPS 42 GD AL B	Apple Watch Series 11 智能	0.665466	apple watch
7	loop size | size | solo | solo l	1	Apple Pencil Tips - 4 pack	Apple Pencil 筆尖 - 4 個裝	0.78802	apple pencil
7	loop size | size | solo | solo l	2	46mm Light Blush Solo Loo	Apple 46 毫米淡胭脂色單圈手	0.556462	apple pencil
7	loop size | size | solo | solo l	3	46mm Light Blush Solo Loo	Apple 46 毫米淡胭脂色單圈手	0.556157	apple pencil
8	usb | cable | usb usb | lightn	1	USB-C to Apple Pencil Adap	Apple USB-C 至 Apple Penci	0.652159	apple pencil
11	strap | the	1	Scosche WatchIt Keychain A	Scosche WatchIt Keychain A	0.709104	apple watch
12	airtag | air | iphone | macboo	1	Smart Folio for iPad (A16) –	Apple 智慧型摺套適用於 iPad	0.627324	蘋果手機
12	airtag | air | iphone | macboo	2	iPad A16 (2025) Wi-Fi 256GE	Apple iPad A16 (2025) Wi-Fi	0.625759	蘋果手機
12	airtag | air | iphone | macboo	3	Logitech Crayon for iPad	Logitech Crayon 適用於 iPad	0.623631	apple pencil

Quality-Ranked Product Lists per Semantic Meaning

cluster	cluster_label	rank	semantic_product	semantic_product_zh	semantic_score	keyword_product	keyword_product_zh	keyword_score
0	苹果 | pcs | fruit | 富士	1	【8pcs】New Zealand / Canada	【8個】紐西蘭/加拿大/小蘋	0.646175	American Snake Fruit/Red Apple	美國蛇果/紅蘋果 (4個)	0.059273
4	苹果 | apples | 水果 | fruit	1	Pollution-Free, Fresh Delivery | 8	無公害種植新鮮直送 8個裝	0.688911	Premium Fuji Apples (4 pcs)	特級富士蘋果(4個)	0.111546
6	band | sport band | 表带 |	1	WATCH S11 GPS 42 GD AL BH SF	Apple Watch Series 11 智	0.665466	Twelve South TimePorter Apple	Twelve South TimePorter	0.156327
7	loop size | size | solo | solo	1	Apple Pencil Tips - 4 pack	Apple Pencil 筆尖 - 4 個裝	0.78802	Apple Pencil Tips - 4 pack	Apple Pencil 筆尖 - 4 個裝	0.501108
7	loop size | size | solo | solo	2	46mm Light Blush Solo Loop - S	Apple 46 毫米淡胭脂色單圈	0.556462	46mm Black Solo Loop - Size 4	Apple 46 毫米黑色單圈手環	0.014108
7	loop size | size | solo | solo	3	46mm Light Blush Solo Loop - S	Apple 46 毫米淡胭脂色單圈	0.556157	46mm Black Solo Loop - Size 0	Apple 46 毫米黑色單圈手環	0.014108
7	loop size | size | solo | solo	4	46mm Light Blush Solo Loop - S	Apple 46 毫米淡胭脂色單圈	0.554454	46mm Black Solo Loop - Size 8	Apple 46 毫米黑色單圈手環	0.014108
7	loop size | size | solo | solo	5	46mm Light Blush Solo Loop - S	Apple 46 毫米淡胭脂色單圈	0.553952	46mm Black Solo Loop - Size 3	Apple 46 毫米黑色單圈手環	0.014108
8	usb | cable | usb usb | ligh	1	USB-C to Apple Pencil Adapter	Apple USB-C 至 Apple Per	0.652159	USB-C to Apple Pencil Adapter	Apple USB-C 至 Apple Per	0.529781
12	airtag | air | iphone | macb	1	Smart Folio for iPad (A16) – Lem	Apple 智慧型摺套適用於 iP	0.627324	Logitech Crayon for iPad	Logitech Crayon 適用於 iP	0.21642

Quality-Ranked Product Lists per Semantic Meaning



CONCLUSION, LIMITATIONS, and Decision Alternatives

Conclusion: Working Semantic Search Prototype

- A fully functional end-to-end semantic search prototype was implemented and evaluated against HKTVmall product data
- The prototype successfully handled both English and Chinese queries which are relevant for Hong Kong's bilingual e-commerce market
- The system demonstrated real-world disambiguation capability (e.g. correctly separating Apple (brand) electronics from apple (fruit) grocery products based purely on query context
- Evaluation across 80+ unique queries confirmed the prototype consistently outperforms traditional keyword search on relevance, cluster diversity, and cross-lingual coverage

Limitations

- 1,000 products per keyword is a ceiling, not the full picture. Clusters may not represent the full distribution of the real search results page.
- Our model covers 20 keywords, which is a small amount compared to all the potential products
- Embeddings are static but product catalogue changes daily. The system requires periodic re-collection and re-embedding to stay current. There is no incremental update mechanism.

Decisions made (and their alternatives)

1. Paraphrase-multilingual-mpnet-base-v2 (not an English-only model)

A 768-dimensional multilingual sentence transformer trained on 50+ languages including Chinese.

2. K-Means (not DBSCAN, not HDBSCAN, not GMM)

K-Means with automatic K selection via elbow + silhouette sweep.

3. TF-IDF for cluster labelling (not GPT summarisation, not centroid decoding)

Fit one TF-IDF vectorizer across all cluster documents, then rank terms per cluster by their mean within-cluster score. Top 4 terms become the label.

THANK YOU